

Probing vision and language models for construction waste material recognition

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ABSTRACT

Motivated by the critical role of automatic sorting in construction waste management, recent advancements have leveraged deep learning's ability to capture powerful features within unimodality-based recognition approaches. However, existing methods remain limited due to reliance on solely image-based datasets, restricting feature expression. To solve this, this paper introduces the VL-CSW dataset by considering both image and text modalities. Next, this paper proposes *ConCLIP*, a vision-and-language model tailored for CSW recognition. *ConCLIP* incorporates a pre-feature interaction network for enhanced modality-specific feature learning and leverages a bidirectional contrastive training paradigm alongside supervised task training to optimize its performance across both modalities. Evaluation on VL-CSW datasets demonstrates the *ConCLIP*'s superiority on CSW material classification task, significantly outperforming strong baselines in most settings. Notably, *ConCLIP* achieves performance improvements of 1.83% and 3.41% compared to unimodality methods in VL-Concrete and VL-Metal classification tasks, respectively, highlighting the efficacy of multi-modality in enhancing automatic sorting system performance.

1. Introduction

The emergence of automatic sorting systems for construction waste (CSW) marks a potential paradigm shift in CSW recycling and management strategies, owing to their demonstrably enhanced efficiency compared to conventional manual methods [1–4]. In contrast, traditional manual CSW sorting methods remain labor-intensive and financially burdensome, resulting in automatic CSW sorting has garnered increasing attention from the research community, especially for the CSW recognition. Driven by recent advances in domains like computer vision and natural language processing (NLP), multi-modal pre-training powered by self-supervised learning has demonstrably enhanced recognition performance across diverse tasks in these fields. While the multi-modal approach shows considerable promise for various prediction tasks, its application to CSW recognition in automatic sorting systems remains nascent. Exemplified by Fig. 1(a), the prevailing paradigm for CSW recognition falls within the realm of unimodal methods. In this paradigm, the input CSW image is ingested by an image encoder - typically a convolutional neural network (CNN) [2,5,6] or a Transformer architecture [7,8] - tasked with extracting multi-scale CSW features through diverse CNN kernels. Subsequently, a dedicated recognition

network (e.g., SVM, decision tree, MLP) leverages these extracted features to generate the corresponding labels for the input CSW image. Building upon this paradigm, Zhu et al. employed traditional machine learning algorithms to extract features and textures from segmented image regions, enabling concrete mix identification [9]. Son et al. further explored the performance of various traditional classifiers for distinguishing construction materials like concrete, steel, and wood. Their findings highlighted the potential of model integration for improved accuracy [10]. Subsequently, Davis et al. pioneered a deep convolutional neural network (CNN)-based algorithm for construction waste material recognition, successfully classifying seven common types, both individually and in mixtures [2]. In addition, Lu et al. demonstrated the potential of semantic segmentation for pixel-level classification of waste components [11]. Chen et al. furthered this effort with a hybrid framework for automatic composition estimation [12]. Recently, Dong et al. introduced a novel Transformer-based model specifically tailored for fine-grained mixtures [8]. These efforts reveal a dominant focus on enhancing recognition accuracy, employing techniques ranging from classic support vector machines to state-of-the-art deep convolutional neural networks and Transformer architectures. While

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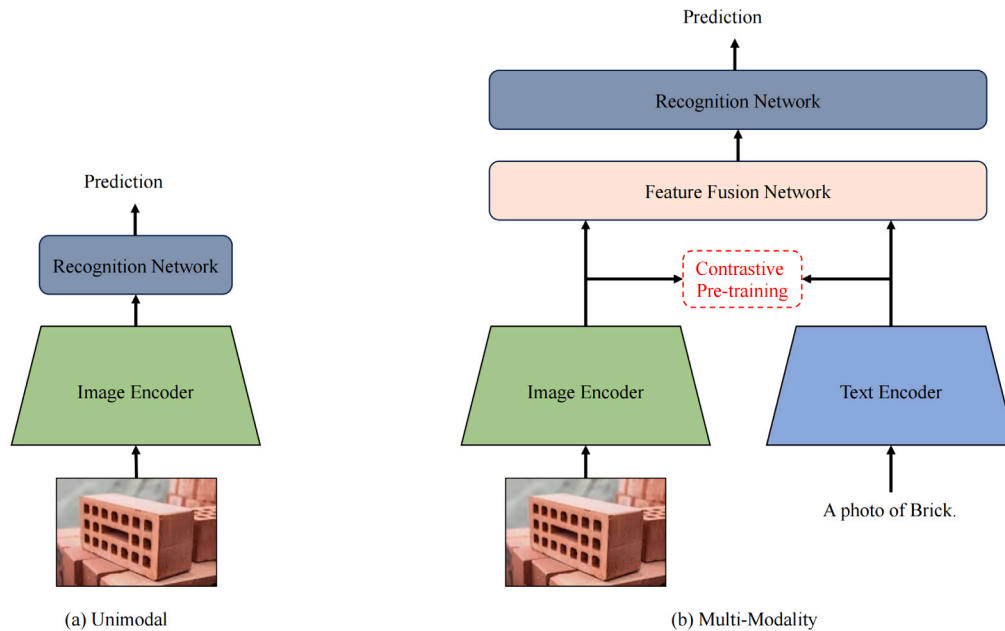


Fig. 1. Unimodal v.s. Multi-modality.

demonstrating promising recognition accuracy, the conventional end-to-end deep convolutional neural network approach exhibits limitations in fully exploiting the distinct features of construction materials during training. This inherent tendency towards learning overly similar material representations presents a significant opportunity for further refining the accuracy of construction material recognition through this method.

The recent surge in multi-modal (Vision and Language) feature mining pipelines, powered by self-supervised learning strategies, has shed light on novel opportunities for performance improvement in image recognition, detection, and semantic segmentation tasks, as exemplified by recent works *CLIP* [13], *ViLT* [14], *BLIP* [15], *Vlmo* [16], etc. As depicted in Fig. 1(b), for a given image x , the prevailing modality strategy relies on generating a textual description (e.g., “a photo of brick”). Subsequently, dedicated image and text encoders process the image and text, respectively, to extract their corresponding feature representations. To enhance alignment between these features, contrastive pre-training (e.g., *CLIP* [13]) is typically employed, minimizing their distance in the latent space. Subsequently, a feature fusion network aggregates the image and text features, enriching the input for the recognition network and thereby boosting prediction performance. This paradigm enables fine-grained material discrimination through text-guided image recognition. Inspired by existing Vision and Language models [13,15–17], this study aims at investigating the vision and language models for construction material recognition. To this end, this paper proposes vision and language model *ConCLIP* to accomplish accurate construction waste material recognition to further optimize performance of the automatic sorting system. Specifically, the proposed model has a similar architecture to the *CLIP* but a modified feature fusion. As a pioneer, *ConCLIP* first constructs the construct waste material multi-modality dataset (i.e., image–text dataset) with the help of high quality construction waste material dataset mentioned in [3]. The image encoder is based on the deep convolutional neural networks and the text encoder is based on Transformer mentioned in BERT [18]. To distinguish *ConCLIP* from *CLIP*, *ConCLIP* further propose a pre-feature fusion strategy that conduct deep feature interaction before contrastive alignment, which promotes the mutual learning of image features and corresponding text features. Finally, the proposal introduces a language modality as a supervision signal for each given image to enhance the representation learning through jointly optimizing the supervised loss

and contrastive loss and thus improve the prediction performance of the construction material recognition. This paper shown that *ConCLIP* outperform the *CLIP* due to the pre-feature fusion strategy, yet with competitive or better construction mater recognition performance. In conclusion, the contributions are summarized as follows:

- **A vision and language model for CSW recognition.** To address the challenge of accurate construction waste material recognition, this paper presents *ConCLIP*, a novel framework that leverages the synergy of vision and language modeling.
- **Vision and Language CSW (VL-CSW) Dataset Construction.** To facilitate the development of vision and language models for construction waste material recognition, this paper established a novel multimodal dataset. Each image of a material was paired with a corresponding textual description, creating image–text pairs that constitute the core of the dataset.
- **Pre-feature Interaction Fusion.** To further enrich image representations, this paper leverages complementary modalities via two encoders: a text encoder capturing features from textual descriptions associated with each image, and an image encoder extracting visual features. This paper then introduce a novel pre-feature interaction fusion (PFIF) framework for interweaving text and image features before contrastive training. This PFIF framework fosters interactive learning between modalities, demonstrably enhancing the resulting representations.
- **Supervised and Contrastive Training (SCT).** To achieve alignment between text and image features, this paper employers a hybrid training strategy. This strategy leverages both supervised and contrastive learning paradigms. Supervised training optimizes feature representations for task-specific prediction accuracy, while contrastive training enforces alignment through direct pairwise matching of text and image features.
- **Empirical Evidence.** This paper conducts extensive experiments on real-world datasets to validate the effectiveness and practicality of our model. The empirical results demonstrate that *ConCLIP* outperforms existing methods consistently on construction waste datasets.

2. Related works

2.1. Waste construction material management

Building upon the comprehensive CDWM efficacy framework proposed by Kabirifar et al. (2021) [19], which integrates stakeholder attitudes and the waste life cycle, this study delves deeper into the influence of reduction, reuse, recycling, and disposal strategies on achieving sustainable CDWM practices. Ghafourian et al. [20] paved the way by integrating sustainability dimensions into CDWM, and this work further illuminates the crucial role of these strategies in optimizing material resource circularity. Recent advancements in CDWM research showcase diverse approaches. Bao et al. [21] present a decision-support framework utilizing qualitative methods like case studies and interviews to achieve cyclic planning for construction waste, exemplified by a case study in Shenzhen, China. Lu et al. [22] leverage a data-driven approach to analyze the volume density of different construction waste streams in Hong Kong, providing valuable insights for resource management optimization. Hoang et al. [23] offer a financial and economic assessment of CDWM recycling in Hanoi, Vietnam, highlighting the cost implications of informal waste disposal methods like landfills. Ma et al. [1] propose an evolutionary game theory-based model for CDWM recycling management, emphasizing the critical role of government incentives for construction waste companies. Their findings reveal that while subsidies can accelerate successful recycling, producer behavior significantly influences the system's dynamics. Notably, increasing landfill costs without subsidies can incentivize cooperation between recyclers and producers. However, existing research has limitations. Long et al. [24] employ an evolutionary game model to analyze producer-recycler interactions under government rewards and punishments, but primarily focus on the dynamic evolution of enterprises, neglecting the government's broader influence on stakeholder behavior. Addressing this gap, Su [25] proposes a three-party evolutionary game framework incorporating government agencies, waste recyclers, and waste producers, revealing the multifaceted roles of government intervention across different phases of CDWM recycling. Du et al. [26] further contribute by analyzing the behavior of government, contractors, and the public using evolutionary game theory, highlighting the effectiveness of incentives and punishments in curbing illegal dumping of construction waste. Overall, this review underscores the significant progress made in CDWM research, while also identifying key areas for future investigation. Moving forward, a deeper understanding of stakeholder interactions, the interplay between economic and environmental factors, and the efficacy of policy interventions will be crucial for optimizing CDWM practices and achieving sustainable waste management.

2.2. Construction waste recognition

Driven by the imperative of green development and sustainability, the efficient recycling and management of construction waste materials emerges as a critical challenge. A pivotal component of this endeavor lies in the accurate classification of diverse waste materials. Previous efforts have primarily relied on manually crafted features and traditional machine learning algorithms, such as Support Vector Machines (SVM) [27] or integrated models [10]. These approaches, however, often struggle to effectively manage the inherent complexity and variability of construction waste materials due to the limitations of manually derived features. In recent years, convolutional neural networks (CNNs) have gained significant traction due to their ability to automatically extract meaningful features through deep learning architectures, offering a promising avenue for construction waste material classification. Several studies have leveraged the power of deep CNNs in this domain, with notable examples including Davis et al.'s work on differentiating seven types of materials, including single and combined types [2], and Chen et al.'s hybrid model integrating physical features

with DenseNet-169 for robust classification [12]. While object detection methods like Faster-R-CNN and Mask-RCNN have demonstrated success in material recovery, their inability to handle multiple waste materials simultaneously restricts their practical applicability [28]. Addressing this limitation, Lu et al. employed DeepLabv3+ semantic segmentation for analyzing combined waste materials, highlighting the potential of this approach for highly unstructured materials, yet requiring further accuracy improvements for real-world deployment [11]. Notably, Dong et al. introduced a Transformer-based boundary-aware semantic segmentation model, showcasing the effectiveness of alternative deep learning architectures like Transformers in accurately delineating material boundaries [8]. However, current methodologies in construction waste identification primarily rely on unimodal data processing, neglecting the potential performance gains offered by exploiting the richness of multimodal datasets. Furthermore, the integration and synergistic effects of vision and language models in this domain remain largely unexplored. To advance the field towards accurate and efficient waste management, future research should prioritize the optimization of both multimodal data utilization and training strategies. This would pave the way for robust construction waste material classification, a crucial step towards sustainable waste management practices.

3. Vision and language CSW datasets

Accurate construction waste material recognition underpins the burgeoning field of automated sorting and sustainable waste management in construction, rendering it a cornerstone of contemporary construction management. While high-quality, labeled datasets have propelled groundbreaking advances in both computer vision (e.g., the transformative impact of foundation models like InternImage [29]) and natural language processing (e.g., GPT-V3 [30]), the field of CSW recognition currently faces hurdles stemming from limitations in existing datasets. This paper identifies these critical issues and argue for their urgent addressal to ensure continued progress in this nascent domain.

3.1. Limitations of existing CSW datasets

Existing construction waste material datasets for model development often fall short of the multi-modal complexity encountered in real-world applications. For instance, commonly used datasets in Refs. [2, 3,10] consist solely of images (Table 1). However, real-world waste materials typically present with at least two modalities, such as image and text, demanding comprehensive analysis. Notably, humans readily recognize such materials through both visual and textual cues; for example, bricks are readily identified both by sight and by descriptive text. Unfortunately, models trained solely on limited modalities, primarily images, often lack generalizability to other modalities, such as language, hindering their practical efficacy.

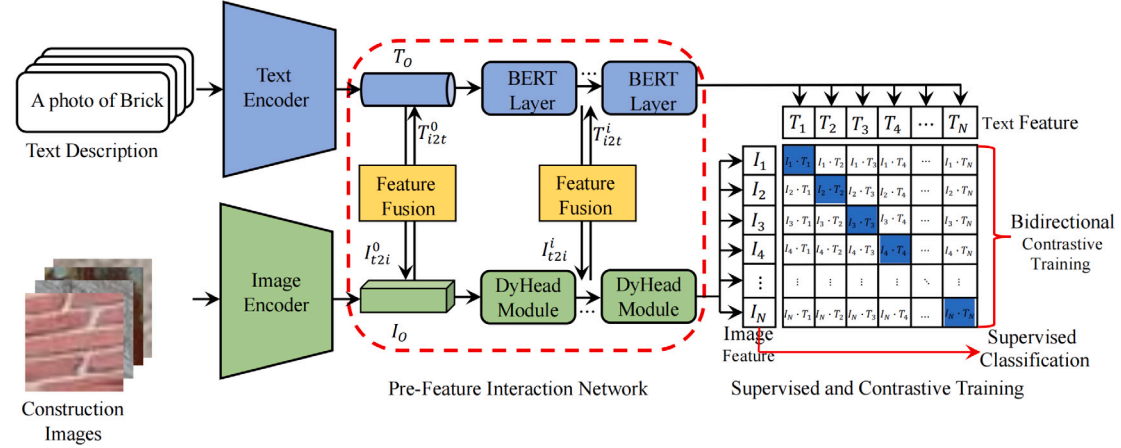
3.2. The vision and language CSW dataset

To formally introduce the proposed vision and language CSW (VL-CSW) dataset, this section delves into its construction and design, specifically focusing on its efficacy in evaluating the accuracy of current and future CSW prediction models under multi-modality scenarios. At its core, the dataset leverages the power of natural language supervision to learn informative features, transcending the standard image classification paradigm that relies solely on named or described categories for post-image classification. The core idea behind text supervision signals, also known as text prompts, is to extract supervising information from natural language to construct image-text pairs for enhancing the extraction of image of the encoder features through learning. This idea is widely applied in supervised, self-supervised, unsupervised, and weakly supervised learning, and has driven the development of multi-modal research in the visual field.

Table 1

Comparisons between multi-modality CSW dataset and other popular datasets. Modality: the number of modality associated with each dataset; Label: the number of CSW categories involving in each dataset; Types: different CSW types including in each dataset.

Released source	Dataset	Modality	Label	Types
Son et al. [10]	CSW Material	1	4	Concrete, Metal, and Wood, Mixture
Davis et al. [2]	CSW Material	1	6	Wood, Plastic, Brick, Concrete, Cardboards, Polystyrene
Sun et al. [3]	CSW Material	1	8	Concrete, Brick, Metal, Wood, Stone, Uncertain, Background, Mixture
ConCLIP (Ours)	CSW Material	2	8	Concrete, Brick, Metal, Wood, Stone, Uncertain, Background, Mixture

**Fig. 2.** Overview of ConCLIP.

However, prevalent data annotation practices, typically represented by numerical IDs and separate category mapping files, often lack the organization and context crucial for effective zero-shot learning. Moreover, text annotations frequently suffer from ambiguity and lack of context, hindering the text encoder's ability to accurately interpret the information. For instance, in ImageNet [31], “construction cranes” and “cranes” refer to distinct entities, highlighting the limitations of single-word labels. Furthermore, images rarely map to single words; instead, they necessitate comprehensive sentence descriptions.

Therefore, this study adopts the format “A photo of a [label], or a construction waste of a [label]” for text sentences, enriching image descriptions and demonstrably improving performance compared to traditional label methods. This prompt-based approach achieves a 1.3% accuracy gain on ImageNet. Notably, while the VL-CSW dataset shares similarities with the one presented in Sun et al. [3], it distinguishes itself by encompassing both image and text modalities, resulting in VL-Concrete, VL-Brick, VL-Metal, VL-Wood, and VL-Stone datasets and totally 105,000 image–text pairs.

4. Approach

Fig. 2 presents the research framework of the multi-modality construction material recognition through jointly supervised and contrastive training, which includes input (images, text pairs), encoders (image encoder, text encoder), image–text feature interactive learning and the training objective. The input module contains the corresponding construction waste materials image and its text description, such as “a photo of [brick]”. Then, the image encoder takes as input an image and returns the corresponding image features, while the text encoder takes as input the text description of the given image and returns the corresponding text feature. Subsequently, a pre-feature interaction fusion network is introduced to interactively learn feature from each modality and thus enhance the learned representation of each modality, thereby effectively improving the accuracy of construction waste material recognition. Finally, the supervised and contrastive training strategy are optimized to further improve the learned representation of each modality and thus improve the prediction accuracy.

4.1. Encoders

4.1.1. Text encoder

To capture representational features of supervised natural language signals, this study leverages the Bidirectional Encoder Representations from Transformers (BERT) model [18]. Specifically, BERT employs a bidirectional Transformer encoder for prompt word encoding, organized into distinct input, encoding, and output layers (Fig. 3). The model architecture incorporates multiple stacked Transformer layers, each comprising multi-head attention mechanisms, residual and normalization layers, and feedforward networks. Notably, the multi-head attention layers leverage an attention mechanism to compute per-layer attention scores for each prompt word, effectively capturing inter-word dependencies and facilitating feature learning.

As shown in Fig. 3, the input layer of the BERT model first converts the input text information into a sequence of word vectors through a linear mapping. Then, the respective position encoding and paragraph encoding are added to the sequence of word vectors. Among them, the position encoding refers to the representation of the position information of each word in the text in vector form, which represents the logical order of the text. The position encoding plays a crucial role in capturing the relationship between different word positions for the text encoding. The paragraph vector is used to handle questions about different texts. Upon securing an appropriate word vector encoding, the model feeds it into the encoding layer, a multi-layered Transformer stack. The Transformer, BERT's core module, hinges on the attention mechanism, which, as the diagram illustrates, is the foundation of BERT's architecture alongside residual and normalization layers and a forward feed neural network. Notably, the multi-head attention mechanism employs three input vectors: $Q(Query)$, $K(Key)$, $V(Value)$. Recognizing that a word's meaning often intertwines with its context, the mechanism leverages the target word as the Q vector, representing context-related words with the $K(Key)$ vector, and weighting them based on the Q and $K(Key)$ similarity. This weighted context is then integrated into the target word vector V , enriching its representation.

The multi-head attention mechanism commences by linearly projecting Q , K , and V matrices into corresponding vector representations.

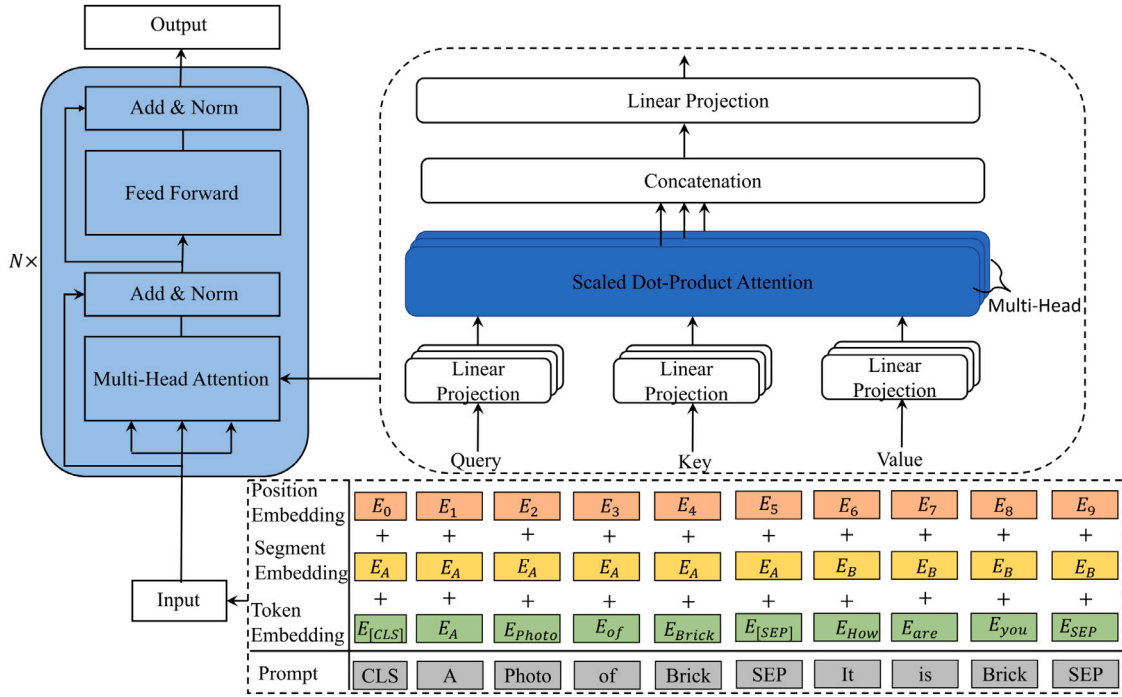


Fig. 3. Transformer model.

Subsequently, it computes the cosine similarity between the $\mathbf{Q}$ vector and each $\mathbf{K}$ vector, yielding the respective attention scores.

$$\alpha_i = \text{softmax} \left(s(\mathbf{K}_i \mathbf{Q}) \right) = \text{softmax} \left(\frac{\mathbf{K}_i^T \mathbf{Q}}{\sqrt{d}} \right) \quad (1)$$

$$\mathbf{H} = \mathbf{V} \text{softmax} \left(\frac{\mathbf{K}^T \mathbf{Q}}{\sqrt{d}} \right) \quad (2)$$

Within this framework, a single attention head leverages the d -dimensional vectors $\mathbf{Q}$ and $\mathbf{K}$ vectors, with $\mathbf{Q} = \mathbf{K}$, to produce the single-head attention output H . In contrast, the multi-head attention mechanism employs numerous information channels parallelly, each governed by its own query vector $\mathbf{Q}$. These dedicated heads selectively attend to different facets of the information, their collective output subsequently spliced to form the final representation. Notably, the Transformer architecture augments this multi-head attention module with a residual connection, layer normalization, and a feed-forward neural network layer. The residual component, inspired by ResNet, mitigates overfitting and vanishing gradients prevalent in deep networks. Meanwhile, the feed-forward network updates the text vector and its associated feature representation [CLS] via linear transformations. This architecture empowers BERT with its defining characteristic: the simultaneous capture of bidirectional context relevance within text.

Based on this, the following text features can be obtained.

$$\mathbf{T} = \text{BERT}(X) \quad (3)$$

where, X represents the corresponding text description of the given CSW image. BERT is a text encoder, which takes as input a text description of given CSW image and returns corresponding feature.

4.1.2. Image encoder

This work introduces an image encoder dedicated to mapping construction waste material images into corresponding feature vectors. Notably, the encoder operates independently of specific deep convolutional neural network (CNN) architectures, readily supporting established image recognition models like VGGNet [32], GoogleNet [33],

and ResNet [34]. This model-agnostic design facilitates the extraction of the following image features

$$\mathbf{O} = \text{ImEncoder}(Y) \quad (4)$$

where, Y represents the input image of construction waste material, processed by the image encoder ImEncoder .

4.2. Pre-feature interaction network

As illustrated in Fig. 4, unlike the approach used in CLIP [13], where direct modality contrast can lead to a feature collapse problem, ConCLIP introduces a pre-interaction network. This modification allows for the learning of interactive representations before engaging in direct modality contrastive pre-training. Such a design potentially enhances feature learning and mitigates the risk of feature collapse. To further refine feature representations for construction waste material classification, this paper introduces a novel pre-feature interaction network (PFIN) following the text and image encoders (Eqs. (3) and (4)). This PFIN exploits a unique cross-modal interaction paradigm, where each modality acts as a supervisory signal for the other. This interactive learning enhances modality-specific feature extraction and enriches representations before inter-modal fusion, leading to demonstrably improved classification performance. The formal definition of the PFIN is as follows:

$$I_{12i}^i, T_{12i}^i = \mathbf{X} - \text{MHA} \left(I^i, T^i \right), \quad i \in \{0, 1, \dots, L-1\} \quad (5)$$

$$I^{i+1} = \text{ImEncoderLayer} \left(I^i + I_{12i}^i \right), \quad I = I^L \quad (6)$$

$$T^{i+1} = \text{BERTLayer} \left(T^i + T_{12i}^i \right), \quad T = T^L \quad (7)$$

where, ImEncoderLayer and the BERTLayer are image encoder and Transformer layers, respectively. L represents the number of fusion layers. T^0 and I^0 are the features returned by the basic image and text encoders, respectively. Cross-modal communication is achieved by the cross-modal multi-head attention module (X-MHA), as shown in formula (5). Thereafter, single-modal fusion updates the image and text features through formulas (6) and (7), respectively. If the context

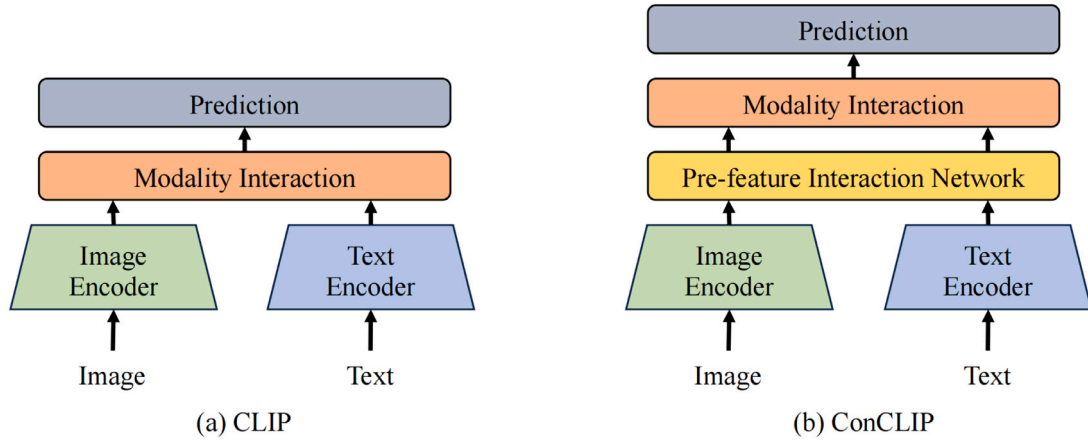


Fig. 4. CLIP v.s. ConCLIP. CLIP directly conducts the modality interaction through contrastive pre-training. While, ConCLIP conducts the modality interaction after pre-interaction network, which are able to learn interactive representations before modality interaction and further enhance the modality feature learning.

vectors (I_{i2i}^i for visual mode and T_{i2i}^i for language mode) are not added, the model degenerates into the late fusion model.

Comprising an image encoder (ImEncoderLayer) and BERT layers (BERTLayer), our multimodal fusion network ingests features from the base image (I^0) and text (T^0) encoders. Intermodal communication is facilitated by a cross-modal multi-head attention module (X-MHA) (Eq. (5)). Subsequent single-modal fusion updates the features through Eqs. (6) and (7). Notably, omitting context vectors (I_{i2i}^i for visual and T_{i2i}^i for language) induces degeneration to a post fusion model.

Within the Cross-modal Multi-head Attention (XMHA) module, each individual head performs an attention operation, calculating an alignment vector from each modality towards the remaining modalities.

$$\begin{aligned} I^{(q)} &= IW^{(q,K)}, T^{(q)} = TW^{(q,L)}, \text{Attn} = I^{(q)} (T^{(q)})^\top / \sqrt{d}, \\ T^{(v)} &= TW^{(v,L)}, I_{i2i} = \text{SoftMax}(\text{Attn}) T^{(v)} W^{(\text{out},K)}, \\ I^{(v)} &= IW^{(v,K)}, T_{i2i} = \text{SoftMax}(\text{Attn}^\top) I^{(v)} W^{(\text{out},L)}, \end{aligned} \quad (8)$$

where, $\{W^{(\text{symbol},K)}, W^{(\text{symbol},L)}, \text{symbol} \in \{q, v, \text{out}\}\}$ are parameters that can be learned.

Pre-feature interaction network offers two salient advantages. First, it unlocks the synergistic potential of image and text features, enabling image classification guided by linguistic cues. This, in turn, paves the way for broader application to diverse recognition tasks like object detection and semantic segmentation. Second, the fusion model embeds image features within the language model, leading to a significant reduction in the laborious and costly process of image annotation.

4.3. Supervised and contrastive training

4.3.1. Supervised training

Building upon the work of Sun et al. [3], this paper employs the Large-Margin Softmax (LSoftmax) function as the supervised loss for our construction waste materials recognition model. This choice facilitates the extraction of task-specific features by virtue of its mathematical formulation, explicitly provided in Eq. (9).

$$\mathcal{L}^{\text{cls}} = p_j^{(i)} = \frac{e^{\|w_j\| \|a^{(i)}\| \psi(\theta_j)}}{e^{\|w_j\| \|a^{(i)}\| \psi(\theta_j)} + \sum_{l \neq j} e^{\|w_l\| \|a^{(i)}\| \cos(\theta_l)}} \quad (9)$$

$$\psi(\theta_j) = (-1)^k \cos(m\theta_j) - 2k, \theta_j \in [k\pi/m, (k+1)\pi/m] \quad (10)$$

where, $k \in [0, m-1]$ is an integer, m controls the margin among classes. When $m = 1$, the L-Softmax loss reduces to the original softmax loss. Through controlled manipulation of the margin parameter m separating classes, the learning objective is demonstrably rendered more rigorous, consequently mitigating the likelihood of overfitting.

4.3.2. Bidirectional contrastive training

Despite its prevalence in supervised deep classification tasks, the cross-entropy loss function has drawn scrutiny for its vulnerability to noise and limitations in mitigating misclassification errors. These shortcomings contribute to suboptimal generalization for models trained with it, even achieving underwhelming performance on benchmark datasets like ImageNet [31]. In response, self-supervised representation learning methods like contrastive training have gained traction across domains including visual, natural language, and graph learning. This paradigm, often leveraging data augmentation and negative sampling, essentially minimizes the distance between a data point and its positive counterpart while maximizing separation from negatives, ultimately aiming to maximize mutual information between samples. Here, this paper proposes investigating contrastive training applied to both image and text features, aiming to bring the target image feature closer to its corresponding text embedding in the latent space while distancing it from negative pairs.

To achieve this, *ConCLIP* embarks on a multi-modal embedding journey, concurrently training an image and text encoder to forge bonds of maximal cosine similarity between image and text representations of N true pairs within each batch, while simultaneously severing ties by minimizing similarity between the $N^2 - N$ incorrect pairings. To guide this process, this paper optimizes a symmetric cross-entropy loss function sculpted from these carefully crafted similarity scores. During training, a minibatch of N input pairs, denoted (x_I, x_T) , is drawn from the training data. Corresponding to each pair, a representation pair (I, T) is computed. This paper denotes the i th pair of representations as (T_i, I_i) . The *ConCLIP* contrastive training objective leverages two distinct loss functions. The first, an image-to-text contrastive loss for the i th pair is defined as:

$$\mathcal{L}_i^{(I \rightarrow T)} = -\log \frac{\exp(\langle I_i, T_i \rangle / \tau)}{\sum_{k=1}^N \exp(\langle I_i, T_k \rangle / \tau)} \quad (11)$$

where $\langle I_i, T_i \rangle$ denotes the cosine similarity, i.e., $\langle I, T \rangle = I^\top T / \|I\| \|T\|$; and $\tau \in \mathbb{R}^+$ is a temperature parameter. This paper employs a loss function that mirrors the InfoNCE loss (Oord et al. [35]), with its minimization driving encoders to uphold maximal mutual information between true pairs within the representational space. Conceptually, this aligns with the log loss of an N -way classifier tasked with predicting the true pair (I_i, T_i) . Notably, in contrast to prior works that leverage contrastive loss within a single modality (Chen et al. [36]; He et al. [37]), our approach distinguishes itself through the adoption of asymmetric image-to-text contrastive loss for each input modality. Consequently, this paper introduces a parallel text-to-image contrastive loss, defined as:

$$\mathcal{L}_i^{(T \rightarrow I)} = -\log \frac{\exp(\langle T_i, I_i \rangle / \tau)}{\sum_{k=1}^N \exp(\langle T_i, I_k \rangle / \tau)} \quad (12)$$

Leveraging weighted summation, this paper constructs the final contrastive training objective from the two independent losses. This composite objective is then averaged across all positive image–text pairs within each minibatch, driving the optimization process.

$$\mathcal{L}^{self} = \frac{1}{N} \sum_{i=1}^N \left(\lambda \mathcal{L}_i^{(I \rightarrow T)} + (1 - \lambda) \mathcal{L}_i^{(T \rightarrow I)} \right) \quad (13)$$

where $\lambda \in [0, 1]$ represents a scalar weight.

4.3.3. Final objective function

To train the image and text encoders in an end-to-end manner described in this study, this paper jointly optimizes the supervised loss L_{cls} and contrastive loss L_{self} . Specifically, the total objective function is defined as shown in Eq. (14).

$$\mathcal{L} = \arg \max \sum_i \cdot \mathcal{L}_i^{self} - \mathcal{L}_i^{cls} \quad (14)$$

where, i represents the training sample in minibatch.

5. Experiments

To rigorously validate the efficacy of the *ConCLIP* architecture proposed herein, this paper conducted a battery of experiments leveraging the newly curated VL-CSW datasets (as detailed in Section 3). This section systematically dissects the experimental design, encompassing the training and testing datasets, the chosen evaluation metric, the image encoder architecture, and the meticulous setup employed. Finally, this paper unveils the compelling results and unveils the *ConCLIP*'s impressive capabilities.

5.1. Datasets

Comprehensive investigations were conducted on five distinct VL-CSW datasets, including VL-Concrete, VL-Brick, VL-Metal, VL-Wood, and VL-Stone datasets. Further validation was performed on the combined VL-CSW dataset to achieve recognition of each primary CSW material.

1. Evaluate the performance on VL-concrete datasets such as Concrete, Background, and Uncertain.
2. Scenario 2: Evaluate the performance on VL-brick datasets such as Brick, Background, and Uncertain.
3. Scenario 3: Evaluate the performance on VL-metal datasets such as Metal, Background, and Uncertain.
4. Scenario 4: Evaluate the performance on VL-wood datasets such as Wood, Background, and Uncertain.
5. Scenario 5: Evaluate the performance on VL-stone datasets such as Stone, Background, and Uncertain.
6. Scenario 6: Evaluate the performance on combined datasets such as VL-Concrete, VL-Brick, VL-Metal, VL-Wood, and VL-Stone.

5.2. Evaluation metric

The model's accuracy was assessed using a confusion matrix, populated with true positive (TP), true negative (TN), false positive (FP), and false negative (FN) values. TP denoted images correctly classified as construction materials, while FP represented instances where background or uncertainty images were misclassified as construction materials. Conversely, FN represented construction images misclassified as background or uncertainty, and TN captured accurate classifications of background or uncertainty images. Performance on the test set was evaluated using precision, recall, accuracy, and F1-score, calculated via the scikit-learn library [38]. The results are presented as follows:

- **Precision:** To assess the reliability of model-based identification of construction elements within visual data, precision is employed. This metric calculates the proportion of images accurately classified as construction material relative to the overall number of such classifications.

$$\text{Precision} = \frac{\sum_{i=1}^N TP_i}{\sum_{i=1}^N TP_i + FP_i} \quad (15)$$

- **Recall (Sensitivity):** The metric of recall quantifies the model's ability to accurately identify all relevant construction images within the sample, expressed as the ratio of correctly classified examples to the total number of construction images in the test set.

$$\text{Recall} = \frac{\sum_{i=1}^N TP_i}{\sum_{i=1}^L TP_i + FN_i} \quad (16)$$

- **Accuracy:** The model's accuracy was evaluated by quantifying its precision, measured as the ratio of true positives to all predicted construction materials within the test set.

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + FN + TN)} \quad (17)$$

- **F1-score:** The F1-score provides a composite evaluation metric, balancing precision and recall by incorporating both false positives (FP) and false negatives (FN). As a weighted average of these measures, it effectively assesses the classification system's accuracy and completeness.

$$\text{F1-Score} = \frac{\sum_{i=1}^N 2TP_i}{\sum_{i=1}^N 2TP_i + FP_i + FN_i} \quad (18)$$

5.3. Image encoder

The proposed framework is encoder-agnostic; therefore a classic deep convolutional image recognition neural network can be selected as the image encoder. Follow the Ref. [3], the VGGNet [32], GoogleNet [33], ResNet [34], and SqueezeNet [39] are selected as image encoder to embed each input image.

Leveraging its image encoder-agnostic nature, the proposed framework readily accommodates a classic deep convolutional image recognition neural network as the image encoder. Building upon the precedent established in Ref. [3], VGGNet [32], GoogleNet [33], ResNet [34], and SqueezeNet [39] were evaluated as candidate image encoders for embedding each input image.

5.4. Experimental setup

This study aims to evaluate and compare the classification performance of construction waste materials by investigating the role of multi-modality, represented by the inclusion (w/L^{self}) or exclusion (w/oL^{self}) of self-learning representations. This paper further optimizes the objective function jointly to leverage multi-modality and maximize classification accuracy. The work of Sun et al. [3] serves as a reference point for the unimodal scenario (i.e., image). All models were trained with a learning rate of $1e-3$, utilizing the Adam optimizer. A batch size of 32 was employed during both training and testing. The experiments were conducted on an Nvidia TITAN-V GPU within a Ubuntu 16.04 environment, using PyTorch 1.4 as the deep learning framework.

5.5. Experiment results

5.5.1. Effects of PFIN and contrastive learning

This paper investigated the effects of Pre-Feature Interaction Networks (PFIN) and bi-directional contrastive learning on a dataset consisting of concrete, brick, metal, wood, and stone materials, using

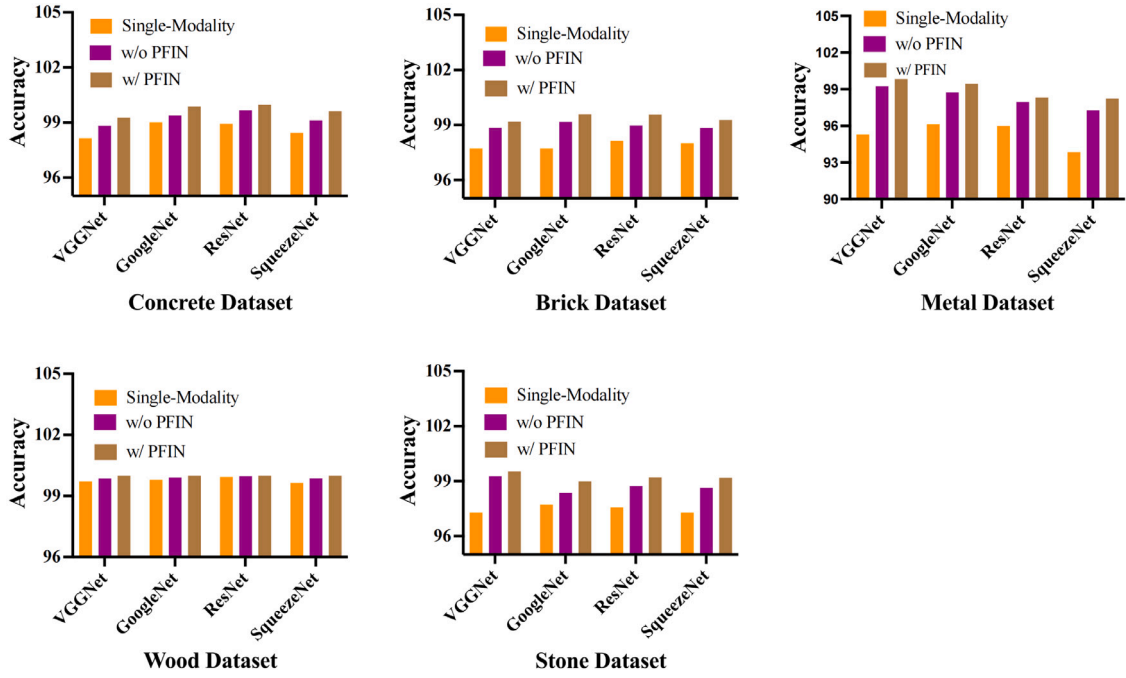


Fig. 5. Effects of PFIN and contrastive learning.

Table 2

Comprehensive evaluation of VL-Concrete datasets.

Loss function	Architectures	Categories	Performance evaluation (%)		Average (%)			
			Accuracy	F1	Accuracy	Precision	Recall	F1
$w/o L^{self}$	VGGNet	Concrete	98.14	99.06	96.59	96.59	96.59	96.58
		Background	96.84	98.40				
		Uncertainty	94.77	97.31				
	GoogleNet	Concrete	99.00	99.50	97.28	97.28	97.28	97.27
		Background	97.27	98.62				
		Uncertainty	95.56	97.73				
	ResNet	Concrete	98.93	99.46	97.37	97.37	97.37	97.37
		Background	97.13	98.54				
		Uncertainty	96.06	97.99				
	SqueezeNet	Concrete	98.43	99.21	96.90	96.90	96.90	96.89
		Background	96.91	98.43				
		Uncertainty	95.34	97.62				
w/L^{self}	VGGNet	Concrete	99.27	99.79	98.40	98.48	98.40	99.12
		Background	98.35	99.34				
		Uncertainty	97.58	98.23				
	GoogleNet	Concrete	99.87	99.95	98.99	99.15	98.99	99.49
		Background	98.34	99.36				
		Uncertainty	98.77	99.17				
	ResNet	Concrete	99.97	99.85	99.13	99.17	99.13	99.47
		Background	99.18	99.69				
		Uncertainty	98.25	98.88				
	SqueezeNet	Concrete	99.62	100.00	99.15	99.20	99.15	99.52
		Background	99.35	99.93				
		Uncertainty	99.47	98.63				

VGGNet, GoogleNet, ResNet, and SqueezeNet, respectively. Our framework was compared against three scenarios: (1) Single-modality, representing the performance achieved by traditional CNN-based models (e.g., VGGNet, GoogleNet) using only image inputs; (2) w/o PFIN, which excludes the pre-feature interaction network (i.e., CLIP); (3) w/ PFIN, incorporating both PFIN and contrastive loss (i.e., ConCLIP). As illustrated in Fig. 5, our findings indicate that multi-modality input enhances performance in waste type classification, with CLIP outperforming the single-modality approach. Moreover, ConCLIP demonstrates superior performance compared to both single-modality and standalone

CLIP, confirming the effectiveness of PFIN. Consistently, all four models produced similar results.

5.5.2. Comprehensive evaluation of VL-Concrete datasets

Leveraging both visual and textual modalities, this paper explored the efficacy of multi-modal contrastive training for concrete material classification on the VL-Concrete dataset. As reported in Table 2, unimodal networks (VGGNet, GoogleNet, ResNet, SqueezeNet) trained solely on the classification objective ($\mathcal{L}^{cls}$) achieved high accuracy for concrete, background, and uncertain classes. GoogleNet (99.00%

Table 3
Comprehensive evaluation of VL-Brick datasets.

Loss function	Architectures	Categories	Performance evaluation (%)		Average (%)			
			Accuracy	F1	Accuracy	Precision	Recall	F1
$w/o \mathcal{L}^{self}$	VGGNet	Brick	97.71	98.84	97.00	97.01	97.00	97.00
		Background	98.14	99.06				
		Uncertainty	95.14	97.52				
	GoogleNet	Brick	97.71	98.84	97.67	97.67	97.67	97.67
		Background	99.00	99.50				
		Uncertainty	96.29	98.12				
	ResNet	Brick	98.14	99.06	97.86	97.86	97.86	97.86
		Background	98.57	99.28				
		Uncertainty	96.86	98.40				
	SqueezeNet	Brick	98.00	98.99	97.71	97.72	97.71	97.71
		Background	98.86	99.43				
		Uncertainty	96.29	98.11				
$w/\mathcal{L}^{self}$	VGGNet	Brick	99.18	99.64	98.94	98.95	98.94	99.35
		Background	99.38	99.77				
		Uncertainty	98.26	98.63				
	GoogleNet	Brick	99.59	99.78	99.45	99.45	99.45	99.64
		Background	99.41	99.64				
		Uncertainty	99.35	99.49				
	ResNet	Brick	99.58	99.91	99.03	99.03	99.03	99.34
		Background	99.14	99.32				
		Uncertainty	98.38	98.79				
	SqueezeNet	Brick	99.27	99.55	99.06	99.06	99.06	99.30
		Background	99.19	99.41				
		Uncertainty	98.71	98.94				

concrete, 97.27% background) excelled for concrete and background, while ResNet (96.06%) led for uncertain. Overall, ResNet emerged as the top performer with an average accuracy of 97.37%.

However, incorporating the contrastive training objective $\mathcal{L}^{self}$ alongside supervised classification objective $\mathcal{L}^{cls}$ further boosted performance by exploiting the rich semantic information within both images and associated text descriptions. Notably, ResNet attained a remarkable 99.97% accuracy for concrete, while SqueezeNet achieved outstanding scores for background (99.35%) and uncertain (99.47%) classes, propelling it to the overall lead with an average accuracy of 99.15%. This represents a significant 1.83% improvement compared to the unimodal best average. These findings demonstrate the effectiveness of multi-modal contrastive training in mining richer semantic features for concrete material classification and highlight its potential for enhancing the performance of related material classification tasks.

5.5.3. Comprehensive evaluation of VL-Brick datasets

This section presents the analysis of experiments conducted on the VL-Brick dataset using the proposed multimodal contrastive training framework. As detailed in Table 3, even without the contrastive training objective $\mathcal{L}^{self}$ and solely relying on the supervised classification objective $\mathcal{L}^{cls}$, classic deep convolutional neural networks (CNNs) like VGGNet, GoogleNet, ResNet, and SqueezeNet can achieve decent performance on the brick classification task. Notably, ResNet attained the highest accuracy for brick material classification (98.14%), while GoogleNet excelled in identifying the background class (99.00%). For the challenging uncertain brick category, ResNet again emerged as the best performer with 96.86% accuracy. Overall, ResNet achieved the highest average accuracy at 97.86%. These results confirm the ability of these established CNNs to recognize bricks, while also highlighting potential for further improvement, particularly in specific category discrimination.

Motivated by this observation, the proposed multimodal contrastive training framework was introduced to enhance the learning performance of these deep CNNs. This framework jointly optimizes the supervised classification objective $\mathcal{L}^{cls}$ and the contrastive training objective $\mathcal{L}^{self}$, allowing for more comprehensive utilization of the semantic information embedded in both image and text modalities. As evident in

Table 3, all four CNNs exhibited significant performance gains across all brick categories when incorporating the multimodal contrastive training objective. This compelling empirical evidence underscores the effectiveness of the proposed vision and language models in leveraging multimodal data for improved brick classification accuracy.

5.5.4. Comprehensive evaluation of VL-Metal datasets

To validate the efficacy of our multi-modal contrastive training framework, this paper conducted experiments on the VL-Metal dataset. Focusing solely on the classification objective $\mathcal{L}^{cls}$, this paper observed that VGGNet, GoogleNet, ResNet, and SqueezeNet could perform metal material classification (Table 4). GoogleNet achieved the highest accuracy (96.14%) for metal identification, while SqueezeNet excelled in classifying background and uncertain classes (99.71% and 92.00%, respectively). Notably, GoogleNet also attained the best overall average classification accuracy (95.52%). Despite achieving some level of classification, VGGNet, GoogleNet, ResNet, and SqueezeNet exhibited underwhelming performance. This suggests that these deep convolutional models struggled to effectively extract and leverage the diverse features inherent in the metal material dataset, ultimately leading to suboptimal classification accuracy.

Leveraging both contrastive and classification losses, ConCLIP effectively harnessed the rich semantic information encoded in both metal images and their corresponding text descriptions. This multimodal synergy markedly enhanced the performance of VGGNet, GoogleNet, ResNet, and SqueezeNet in metal material classification. Notably, VGGNet attained a peak accuracy of 99.83% for metal materials, while SqueezeNet and GoogleNet excelled in classifying background and uncertain categories, achieving 100.00% and 97.38% accuracy, respectively. GoogleNet emerged as the overall champion, averaging 98.78% accuracy, representing a 3.41% improvement over the best average attainable without the multimodal contrastive objective. This suggests that the inclusion of language supervision signals empowered the selected deep CNNs to exploit the subtle features of similar classes within the metal dataset, ultimately leading to a substantial boost in classification accuracy for both metal materials and related categories.

Table 4
Comprehensive evaluation of VL-Metal datasets.

Loss function	Architectures	Categories	Performance evaluation (%)		Average (%)			
			Accuracy	F1	Accuracy	Precision	Recall	F1
$w/o L^{self}$	VGGNet	Metal	95.29	97.59	94.91	94.96	94.91	94.90
		Background	99.00	99.50				
		Uncertainty	90.43	94.97				
	GoogleNet	Metal	96.14	98.03	95.52	95.59	95.52	95.52
		Background	99.14	99.57				
		Uncertainty	91.29	95.44				
	ResNet	Metal	96.00	97.96	95.24	95.30	95.24	95.23
		Background	99.14	99.57				
		Uncertainty	90.57	95.05				
	SqueezeNet	Metal	93.86	96.83	95.19	95.18	95.19	95.18
		Background	99.71	99.86				
		Uncertainty	92.00	95.83				
w/L^{self}	VGGNet	Metal	99.83	100.00	98.46	98.46	98.46	99.02
		Background	99.49	99.78				
		Uncertainty	96.13	97.29				
	GoogleNet	Metal	99.44	99.89	98.78	98.80	98.78	99.21
		Background	99.53	99.62				
		Uncertainty	97.38	98.11				
	ResNet	Metal	98.33	98.97	97.69	97.70	97.69	98.35
		Background	99.51	99.73				
		Uncertainty	95.23	96.35				
	SqueezeNet	Metal	98.23	99.14	98.47	98.48	98.47	99.11
		Background	100.00	100.00				
		Uncertainty	97.19	98.21				

Table 5
Comprehensive evaluation of VL-Wood datasets.

Loss function	Architectures	Categories	Performance evaluation (%)		Average (%)			
			Accuracy	F1	Accuracy	Precision	Recall	F1
$w/o L^{self}$	VGG16	Wood	99.71	99.86	97.89	97.89	97.89	97.89
		Background	97.08	98.52				
		Uncertainty	96.87	98.41				
	GoogleNet	Wood	99.79	99.90	97.93	97.94	97.93	97.93
		Background	98.01	98.99				
		Uncertainty	96.01	97.97				
	ResNet	Wood	99.93	99.96	98.15	98.15	98.15	98.15
		Background	97.08	98.52				
		Uncertainty	97.44	98.70				
	SqueezeNet	Wood	99.64	99.82	97.67	97.67	97.67	97.67
		Background	97.08	98.52				
		Uncertainty	96.30	98.11				
w/L^{self}	VGGNet	Wood	100.00	100.00	99.68	99.70	99.68	99.90
		Background	100.00	100.00				
		Uncertainty	99.04	99.71				
	GoogleNet	Wood	100.00	100.00	99.11	99.11	99.11	99.57
		Background	99.27	99.81				
		Uncertainty	98.09	98.91				
	ResNet	Wood	100.00	100.00	99.41	99.41	99.41	99.65
		Background	99.17	99.52				
		Uncertainty	99.06	99.43				
	SqueezeNet	Wood	100.00	100.00	99.71	99.73	99.71	99.95
		Background	100.00	100.00				
		Uncertainty	99.14	99.84				

5.5.5. Comprehensive evaluation of VL-Wood datasets

Similar to the other three construction waste material datasets, this paper also conducts the experiments on VL-Wood dataset. As seen in Table 5, when only considering the classification objective function L^{cls} , VGGNet, GoogleNet, ResNet and SqueezeNet demonstrate good classification performance on the VL-Wood material dataset. Particularly, for the classification of wood materials, ResNet achieved the best classification results with 99.93% recognition rate, which is mainly due to the more obvious texture features of wood compared to that of other materials. For other categories in the VL-Wood material dataset, GoogleNet achieved the best results in background classification with

a 98.01% accuracy rate, while ResNet achieved the best results in the uncertain category with a 97.44% accuracy rate.

Following the evaluation protocol established for the other construction waste material datasets, this paper explored the performance of various deep learning models on the VL-Wood dataset. As detailed in Table 5, when focusing solely on classification accuracy (measured by the objective function L^{cls} , VGGNet, GoogleNet, ResNet, and SqueezeNet all exhibited strong performance on the VL-Wood dataset). Notably, ResNet achieved the highest classification accuracy for wood materials with a remarkable 99.93% recognition rate. This likely stems from the more pronounced texture features inherent in

Table 6
Comprehensive evaluation of VL-Stone datasets.

Loss function	Architectures	Categories	Performance evaluation (%)		Average (%)			
			Accuracy	F1	Accuracy	Precision	Recall	F1
w/o L^{self}	VGGNet	Stone	97.29	98.62	97.81	97.81	97.81	97.81
		Background	98.71	99.35				
		Uncertainty	97.43	98.70				
	GoogleNet	Stone	97.71	98.84	98.33	98.34	98.33	98.33
		Background	99.14	99.57				
		Uncertainty	98.14	99.06				
	ResNet	Stone	97.57	98.77	98.10	98.10	98.10	98.09
		Background	99.14	99.57				
		Uncertainty	97.57	98.77				
	SqueezeNet	Stone	97.29	98.62	98.00	98.00	98.00	98.00
		Background	99.29	99.64				
		Uncertainty	97.43	98.70				
w/ L^{self}	VGGNet	Stone	99.53	99.91	99.22	99.24	99.22	99.48
		Background	99.17	99.65				
		Uncertainty	98.95	99.18				
	GoogleNet	Stone	98.99	99.49	99.40	99.41	99.40	99.72
		Background	100.00	100.00				
		Uncertainty	99.22	99.67				
	ResNet	Stone	99.21	100.00	99.46	99.48	99.46	99.85
		Background	100.00	100.00				
		Uncertainty	99.18	99.54				
	SqueezeNet	Stone	99.19	99.95	99.42	99.42	99.42	99.84
		Background	100.00	100.00				
		Uncertainty	99.07	99.57				

wood compared to other materials in the dataset. For other categories within the VL-Wood dataset, GoogleNet emerged as the top performer in background classification with a 98.01% accuracy rate, while ResNet secured the highest accuracy (97.44%) for the uncertain category. Among tested architectures, ResNet emerged as the champion, achieving a stellar average classification accuracy of 98.15%. In comparison, when leveraging both classification and contrastive training objectives, *ConCLIP* efficaciously exploits the semantic resonance between wood and textual descriptions. Intriguingly, the inclusion of the contrastive objective propelled VGGNet, GoogleNet, ResNet, and SqueezeNet to near-perfect performance on the wood category, all attaining 100.00% accuracy. Moreover, VGGNet and SqueezeNet replicated this feat for the background category. For the uncertain category, SqueezeNet reigned supreme with an impressive 99.14% accuracy. In a crowning achievement, SqueezeNet secured the overall accuracy throne, with a remarkable 99.71%, a 1.59% boost over the best unimodal performance. These findings underscore the efficacy of incorporating language modalities to unlock rich semantic features and, consequently, elevate classification accuracy across diverse categories within the wood dataset.

5.5.6. Comprehensive evaluation of VL-Stone datasets

This paper further validated *ConCLIP*'s performance on the VL-Stone dataset (see Table 6). Focusing on the classification objective function $\mathcal{L}^{cls}$, analysis of Table 6 reveals that VGGNet, GoogleNet, ResNet, and SqueezeNet exhibit comparable capabilities in recognizing stones, backgrounds, and uncertain categories. Notably, GoogleNet achieved the highest stone material classification accuracy with 97.71%, SqueezeNet excelled in background recognition with 99.29%, and GoogleNet again dominated the uncertain category with 98.14%.

Among unimodal models, GoogleNet emerged as the top performer, achieving an average classification accuracy of 98.33%. Notably, *ConCLIP* leveraged both supervised classification and contrastive learning objectives, effectively exploiting the semantic features encoded within both image and text information. This dual-objective approach demonstrably augmented the classification accuracy of VGGNet, GoogleNet, ResNet, and SqueezeNet across diverse stone types within the dataset. Specifically, for stone material classification, VGGNet reigned supreme

with an accuracy of 99.53% for the “stone” category. For background classification, GoogleNet, ResNet, and SqueezeNet achieved perfect accuracy (100.00%). Finally, GoogleNet outperformed all other models in the uncertain category, securing an impressive accuracy of 99.22%. Remarkably, ResNet emerged as the most accurate classifier under the multi-modal setting, achieving an average score of 99.46%. Notably, this performance represents a 1.15% improvement over the best unimodal result, highlighting the efficacy of incorporating language modalities.

5.5.7. Comprehensive evaluation of VL-Combined datasets

To further elucidate the comparative performance of our approach, this paper not only evaluated its efficacy on individual datasets for concrete, brick, metal, wood, and stone (VL-Concrete, VL-Brick, etc.), but also conducted experiments on a unified VL-Combined dataset encompassing these main categories. Importantly, the VL-Combined dataset preserved the inherent characteristics of each material category to facilitate targeted analysis. As detailed in Table 7, when only considering the classification objective function, deep CNN models like VGGNet, GoogleNet, ResNet, and SqueezeNet demonstrated strong performance across individual material categories within the combined dataset. Notably, VGGNet, GoogleNet, and SqueezeNet achieved a 99.86% classification accuracy for concrete, surpassing other models. SqueezeNet excelled in brick classification with a perfect 100% accuracy. For metal, GoogleNet and ResNet tied for first place with a 99.71% accuracy. The triumvirate of VGGNet, GoogleNet, and SqueezeNet again dominated wood classification, reaching a 99.71% accuracy. Finally, ResNet claimed the top spot for stone classification with a 99.71% accuracy. Remarkably, GoogleNet emerged as the overall champion for the combined dataset, achieving an impressive average classification accuracy of 99.66%.

While all four models (VGGNet, GoogleNet, ResNet, and SqueezeNet) achieved perfect (100.00%) classification accuracy for the mixture soil and brick, GoogleNet and ResNet attained the same feat for metal and wood. ResNet and SqueezeNet further demonstrated flawless identification of stone materials. Consequently, ResNet emerged as the top performer with an absolute average accuracy of 100.00%, signifying its ability to distinguish all material types within the combined VL dataset. Importantly, compared to unimodal approaches, this

Table 7
Comprehensive evaluation of VL-Combined datasets.

Loss function	Architectures	Categories	Performance evaluation (%)		Average (%)			
			Accuracy	F1	Accuracy	Precision	Recall	F1
$w/o L^{self}$	VGGNet	Concrete	99.86	99.93	99.60	99.60	99.60	99.60
		Brick	99.86	99.93				
		Metal	99.57	99.79				
		Wood	99.71	99.86				
		Stone	99.00	99.50				
	GoogleNet	Concrete	99.86	99.93	99.66	99.66	99.66	99.66
		Brick	99.86	99.93				
		Metal	99.71	99.86				
		Wood	99.71	99.86				
		Stone	99.14	99.57				
	ResNet	Concrete	99.57	99.86	99.46	99.46	99.46	99.46
		Brick	99.71	99.86				
		Metal	99.71	99.86				
		Wood	99.57	99.79				
		Stone	99.71	99.35				
	SqueezeNet	Concrete	99.86	99.93	99.60	99.60	99.60	99.60
		Brick	100.00	100.00				
		Metal	99.57	99.79				
		Wood	99.71	99.86				
		Stone	98.86	99.43				
w/L^{self}	VGGNet	Concrete	100.00	100.00	99.83	99.84	99.83	99.97
		Brick	100.00	100.00				
		Metal	99.79	99.91				
		Wood	99.93	100.00				
		Stone	99.43	99.96				
	GoogleNet	Concrete	100.00	100.00	99.91	99.92	99.91	99.98
		Brick	100.00	100.00				
		Metal	100.00	100.00				
		Wood	100.00	100.00				
		Stone	99.53	99.89				
	ResNet	Concrete	100.00	100.00	100.00	100.00	100.00	100.00
		Brick	100.00	100.00				
		Metal	100.00	100.00				
		Wood	100.00	100.00				
		Stone	100.00	100.00				
	SqueezeNet	Concrete	100.00	100.00	99.97	99.98	99.97	99.99
		Brick	100.00	100.00				
		Metal	99.87	99.94				
		Wood	100.00	100.00				
		Stone	100.00	100.00				

multi-modal strategy yielded a noteworthy 0.34% improvement in performance. These findings underscore the effectiveness of incorporating multi-modality scenarios for comprehensively capturing image and text features of diverse construction waste materials, ultimately leading to enhanced classification accuracy.

6. Discussion

In this phase, this paper aims to discuss the unified evaluation framework for different waste types. Compared with other single-metric evaluations (e.g., accuracy alone), the confusion matrix provides a richer, more nuanced view of model performance. This makes it particularly valuable in real-world applications where understanding the behavior of a classification model in detail is crucial for making informed decisions about its deployment and further development. Therefore, this paper presents the confusion matrices for VGGNet and ResNet applied to our VL-Combined Datasets to provide a clearer understanding of the models' performance. As depicted in Fig. 6, these matrices facilitate the identification of true positives, false positives/negatives, and other outcomes. With this information, this paper can calculate various evaluation metrics such as precision, recall, and F1-score. Thus, by using confusion matrices as our unified evaluation metric, this paper simplifies the understanding of model performance for others.

This paper also introduces the Weighted Accuracy (WA) for the varying complexity of different waste types. According to the results

from Table 7, this paper first calculates the average accuracy for each class using four different models. Next, this paper determines the ratio of each class based on these average accuracies. Accordingly, this paper assigns complexity weights to concrete, brick, metal, wood, and stone as [0.1, 0.1, 0.2, 0.1, 0.5], reflecting the relative difficulty of classification. Consequently, the average accuracy of VGGNet using a traditional approach is 99.60%. However, when incorporating the classification difficulty through weighted accuracy, the result improves to 99.67%. This demonstrates that considering weighted accuracy is crucial in scenarios with imbalanced classes and varying recognition difficulties.

In addition, as shown in Fig. 6, although confusion matrix provides detailed information on true positives, false positives/negatives, etc., offering deeper insights into the classification errors across different waste types. For example, consider the stone classification using VGGNet, with a total of 1400 testing samples: there are instances where stone is mistakenly recognized as concrete (5 cases), brick (1 case), and metal (2 cases). This analysis helps us identify specific causes of accuracy loss. Although the multi-modality method has improved the model's overall performance, there is significant room for further enhancement.

7. Conclusions and future work

To address the lack of fully investigated data modalities in construction waste management, this paper introduces ConCLIP, a vision and

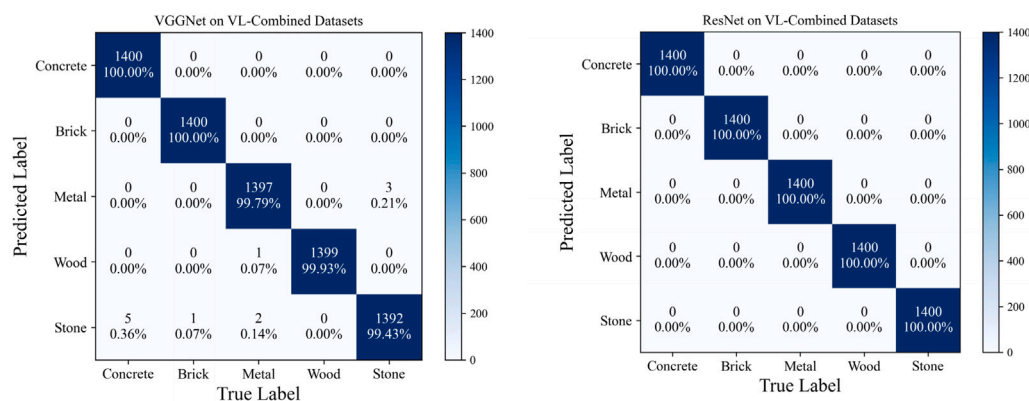


Fig. 6. Unified evaluation metric.

language model for material recognition. *ConCLIP* leverages the CSW dataset, comprising paired high-resolution construction waste images and descriptive text annotations. Key to its success is a bidirectional contrastive objective with supervised task co-training, aligning image representations with their corresponding text descriptions. This strategy surpasses competitive in-domain initialization methods, yielding powerful multimodal representations and compelling recognition results. Notably, *ConCLIP* achieves a 1.83% improvement in concrete recognition and a 3.41% boost in metal classification. Extending beyond image–text pairs, *ConCLIP*’s innovative cross-modal training framework paves the way for incorporating additional data modalities, such as sensor data or material composition reports, crucial for advancing automated construction waste management. This paper anticipates *ConCLIP* to inspire future research towards making more efficient use of multimodal data for sustainable waste management solutions. In the future, this paper plans to investigate prompt engineering more closely, aiming to provide more accurate descriptions of each waste type to reduce the incidence of false positives and negatives during prediction.

CRedit authorship contribution statement

Ying Sun: Writing – original draft, Methodology. **Zhaolin Gu:** Writing – review & editing, Conceptualization. **Sean Bin Yang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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